Level 1 Functional Skills Mathematics
SAMPLE PAPER 2

A City & Guilds Group Business

Version 1.0

Duration: 25 minutes
Total marks: 15

MODEL SOLUTION SCORE
15 / 15 marks (100% Pass)
25 minutes · Verified by MFS

SECTION 1 – CALCULATOR NOT PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

Assessment date (DDMMYYYY): 20102026

Centre number: MFS0001

Candidate signature and declaration*: Master Functional Skills

***I declare that I had no prior knowledge of the questions in this assessment and that I will not share information about the questions.**

Please check that your name is correctly printed on the candidate barcode label. If not, please tell the invigilator before the start of the exam.

You should have the following for this assessment:

- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler
- a protractor.



You must NOT use a calculator for Section 1.

General instructions

- Read through each question carefully.
- Write all your answers in this booklet.
- Check your calculations and check that your answers make sense.

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SECTION 1 – CALCULATOR NOT PERMITTED

There are **15** marks available in this section.

You should check all your work as you go along.

You must **not** use a calculator in this section.



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Q1

846.7 ÷ 100 =

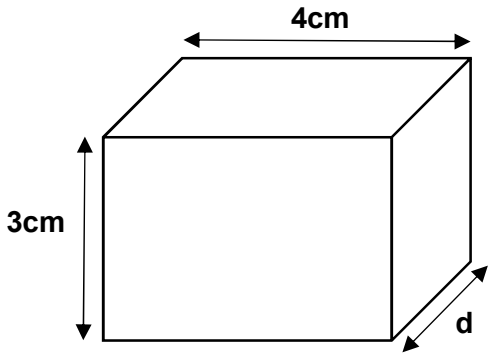
• Dividing by 100 moves decimal point 2 places left:
846.7 ÷ 100 = 8.467

8.467

[1 mark]

(1 mark)

Q2 The volume of this cuboid is 24cm³.



What is the value for the depth d?

• Volume = length x height x depth
4 x 3 x d = 24 --> 12 x d = 24 --> d = 24 ÷ 12 = 2 cm

2 cm

[1 mark]

(1 mark)

Q3

9^2 =

• Square number: 9 x 9 = 81

81

[1 mark]

(1 mark)

Q4

18 – 5 x 2 =

• BIDMAS rule: Multiplication before subtraction:
5 x 2 = 10 --> 18 – 10 = 8

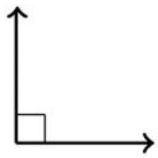
8

[1 mark]

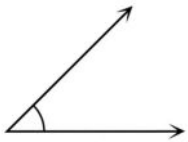
(1 mark)

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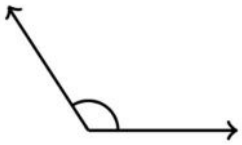
Q5



A



B



C



D

Which of the above has an angle of 180°?

(tick one box)

- **A** ☐
- **B** ☐
- **C** ☐
- **D** ☑ *Option D is correct: 180°*

• Shape D is a straight line, which measures 180°

(A = 90° right angle, B = acute < 90°, C = obtuse > 90°)

[1 mark]

(1 mark)

Q6

Write two hundred and forty-six thousand eight hundred and five in figures

• 246,000 + 800 + 5 = 246,805

246 805

[1 mark]

(1 mark)

Q7

What is $\frac{2}{5}$ as a decimal?

(tick one box)

- **A** 0.4 ☑ *Option A is correct: 0.4*
- **B** 0.2 ☐
- **C** 0.25 ☐
- **D** 0.04 ☐

• 2/5 = 4/10 = 0.4

[1 mark]

(1 mark)

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Q8

− 6 + 4 =

(tick one box)

A	−10	☐
B	− 2	☑
C	2	☐
D	10	☐

• Start at -6 on number line and add 4: -6 + 4 = -2

Option B is correct: -2

[1 mark]

(1 mark)

Q9

What is the range of these weights?

40 kg 65kg 27kg 84kg 34kg

• Range = Highest − Lowest = 84 kg − 27 kg = 57 kg

57 kg

[1 mark]

(1 mark)

Q10 At a rugby match 19 417 home fans and 2 648 away fans attend.

What is the total attendance, rounded to the nearest hundred?

• Total = 19,417 + 2,648 = 22,065

• Round to nearest hundred: 22,065 --> 22,100 (tens digit 6 rounds up)

22 100

[1 mark]

(1 mark)

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Q11 A company conducts a survey. The survey asks people their ages.
The data will be grouped.

• Complete 10-year group intervals consistently:

Complete the table with the missing age groups.

Age group boundaries
25 and under
26 - 35
36 - 45
46 - 55
56 - 64
65 and over

[1 mark]

(1 mark)

Q12 A manager wants to know the ratio of men to women in their factory. There are 135 men and 270 women working at the factory.

What is the ratio of men to women?

• Ratio Men : Women = 135 : 270 [M1]

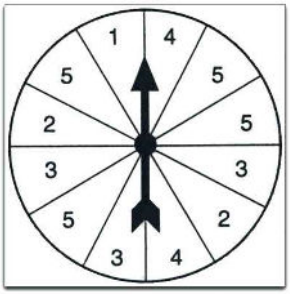
• Divide both parts by 135: 135÷135 = 1, 270÷135 = 2 --> 1 : 2 [A1]

1 : 2

[2 marks]

(2 marks)

Q13 A fair has a spin the arrow game.
A player wins a prize if the arrow lands on a 5.



What is the probability of a player winning a prize?
Write your answer as a fraction in its simplest form.

• Count sections on spinner: Total = 12 sections [M1]

• Sections containing number 5: 4 sections [M1]

• Probability = 4/12 --> Simplify: 4/12 = 1/3 [A1]

Answer: 1/3

[2 marks]

(2 marks)

End of Section 1

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Level 1 Functional Skills Mathematics
SAMPLE PAPER 2

A City & Guilds Group Business
Version 1.0

Duration: 1 hour 20 minutes
Total marks: 45

MODEL SOLUTION SCORE
45 / 45 marks (100% Pass)
1 hr 20 mins · Verified by MFS

SECTION 2 – CALCULATOR PERMITTED

Candidate name (first, last)

First: MASTER

Last: FUNCTIONAL SKILLS

Candidate enrolment number: MFS4748

Date of birth (DDMMYYYY): 01012000

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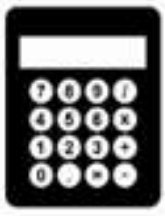
• If you have used any additional answer sheets write the number of additional sheets in this box. ☐
• Please ensure that you **staple** additional answer sheets to the **back** of this booklet, clearly labelling them with your full name, enrolment number, centre number and date in BLOCK CAPITALS.
• You must use a black or blue pen. You may use a pencil for charts and diagrams.

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- a calculator
- a pen with black or blue ink
- a pencil (for diagrams, graphs and charts only)
- an eraser
- a 30cm ruler
- a protractor.



General instructions

- Read through each question carefully.
- Show your working out (where required).
- Write all your working out and answers in this booklet.
- Check your calculations and check that your answers make sense.
- There are additional pages **including graph paper** at the back of this booklet if you run out of space or ask the invigilator if you need additional sheets of paper.

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SECTION 2 – CALCULATOR PERMITTED

There are **45** marks available in this section.

You should check all your work as you go along.

You may use a calculator.



Q1

15	24	16	18	21

What is the mean of these numbers?

- Sum = 15 + 24 + 16 + 18 + 21 = 94
- Mean = 94 ÷ 5 = 18.8

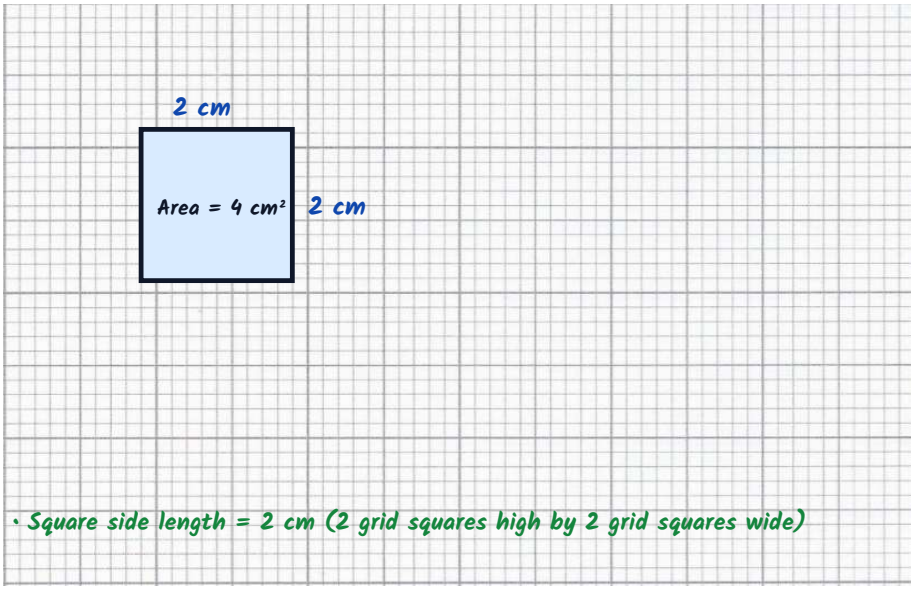
18.8

[1 mark]

(1 mark)

Q2

Draw a square with sides of 2cm



- Square side length = 2 cm (2 grid squares high by 2 grid squares wide)

[1 mark]

(1 mark)

Q3

Write $\frac{33}{8}$ as a mixed number.

- 33 ÷ 8 = 4 with remainder 1 --> 4 1/8

4 1/8

[1 mark]

(1 mark)

Q4

Which of the following lists is in increasing order from smallest to largest?

(tick one box)

• 50% < 52% < 53% < 55%

A	52%	53%	50%	55%	☐
B	50%	52%	53%	55%	☑ Option B is correct: 50% 52% 53% 55%
C	55%	52%	53%	50%	☐
D	50%	53%	52%	55%	☐

[1 mark]

(1 mark)

Q5

140 x _____ = 8400

• Inverse operation: 8400 ÷ 140 = 60

60

[1 mark]

(1 mark)

Q6 A student states that 540cm is the same as 54m.

Are they correct? Explain your answer.

Are they correct? *(tick one box)* **Yes** ☐ **No** ☑

Explanation

Explanation: There are 100 cm in a metre, so 540 cm = 5.4 m.

The student divided by 10 instead of 100. 54 metres would be 5,400 cm.

[1 mark]

(1 mark)

Q7 The formula below can be used to calculate interest on a savings account after one year:

amount put into account x interest rate **as a decimal**

A woman puts £2 500 into an account. The interest rate is 5%.

What will be in the account after one year?

Show your working

Step 1: Convert interest rate to decimal: 5% = 0.05 [M1]

Step 2: Calculate interest: £2,500 × 0.05 = £125.00 interest [M1]

Step 3: Total in account after 1 year: £2,500 + £125 = £2,625 [A1]

£ 2 625 [3 marks]

(3 marks)

Q8 A taxi company charges £5 standing charge and £1.50 per mile for a journey.

A man travels in a taxi for 12.2 miles.

What is the cost of his journey?

Show your working

Step 1: Calculate mileage charge: 12.2 miles × £1.50 = £18.30 [M1]

Step 2: Add standing charge: £18.30 + £5.00 = £23.30 [A1]

£ 23.30 [3 marks]

(3 marks)

Q9 A student receives two loans. One is for £9 153 and the other is for £8 944.

Approximately $\frac{1}{2}$ of the loans will be spent on tuition fees and $\frac{1}{4}$ of the loans will be spent on accommodation.

Estimate the amount of money they will have left.

Show your working

Step 1: Round each loan to convenient numbers (nearest thousand):

£9,153 ≈ £9,000 and £8,944 ≈ £9,000 [M1]

Estimated total loans = £9,000 + £9,000 = £18,000 [M1]

Step 2: Calculate total fraction spent on tuition (1/2) and accommodation (1/4):

Total fraction spent = 1/2 + 1/4 = 2/4 + 1/4 = 3/4 --> Remaining = 1/4

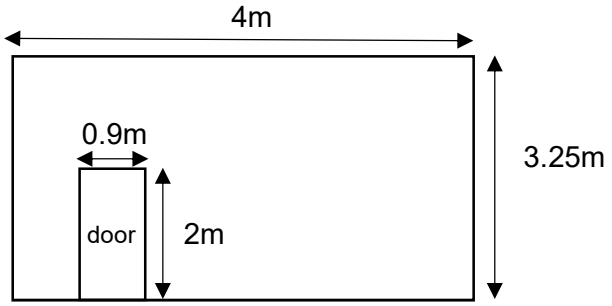
Estimated money left = 1/4 × £18,000 = £18,000 ÷ 4 = £4,500 [A2]

£ 4 500

[4 marks]

(4 marks)

Q10 A decorator is painting a room and has one wall left to paint.
They have enough paint to cover 12m². The dimensions of the wall are shown below:



Sketch not to scale

She will not paint over the door.

Do they have enough paint for the wall?

Explain your answer. Use figures in your answer.

Show your working

Step 1: Calculate total area of wall: 4 m × 3.25 m = 13 m² [M1]

Step 2: Calculate area of door: 0.9 m × 2 m = 1.8 m² [M1]

Step 3: Calculate area left to paint: 13 m² − 1.8 m² = 11.2 m² [A1]

Step 4: Compare with paint supply: 11.2 m^2 < 12 m^2 --> Enough paint! [A1]

Do they have enough paint? *(tick one box)* **Yes** ☑ **No** ☐

Explanation

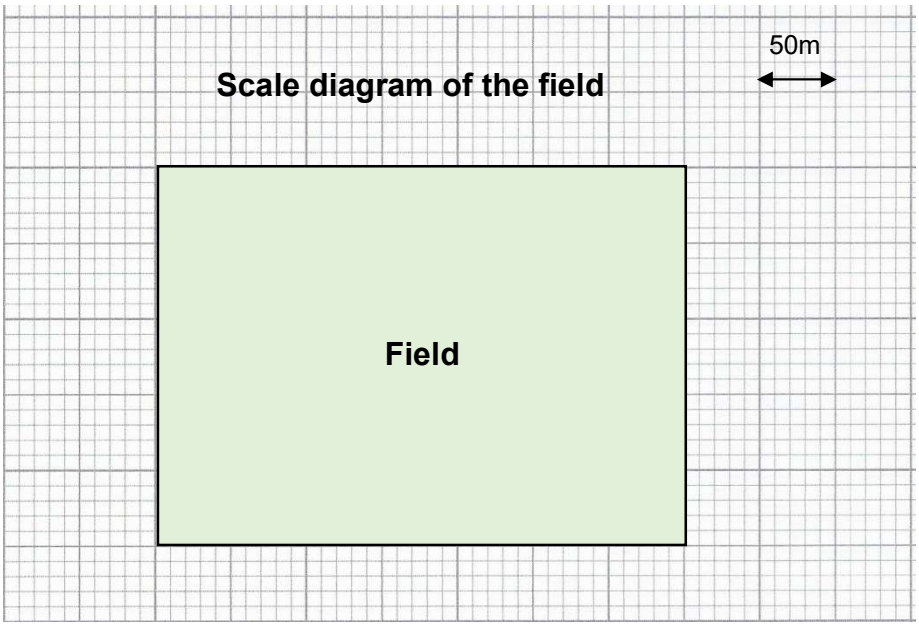
Explanation: The area of the wall to paint is 11.2 m² (13 m² − 1.8 m²).

Since she has enough paint for 12 m², she has enough paint with 0.8 m² left over.

[4 marks]

(4 marks)

Q11 A local sports club organises a sponsored walk around the edge of a field.



Each person will walk round the field 5 times.

How far will each person walk in kilometres?

Show your working

Step 1: Find dimensions of field using grid scale (each square = 50 m):

Length = 7 squares × 50 m = 350 m [M1]

Width = 5 squares × 50 m = 250 m [M1]

Step 2: Calculate perimeter of 1 lap: (350 + 250) × 2 = 1,200 m = 1.2 km [A1]

Step 3: Calculate total distance for 5 laps: 5 × 1.2 km = 6.0 km (or 6,000 m) [A1]

6 km *[4 marks]*

(4 marks)

Q12 A cricket club sells 5,062 tickets for a charity match. 3,484 are Adult tickets and the rest are Under 18s tickets.

Ticket Prices	
Adults	£18
Under 18s	£12

85% of ticket sales pay for the event. The rest goes to community projects.

How much does the club give to community projects?

Show your working

Step 1: Calculate adult ticket revenue: 3,484 × £18 = £62,712 [M1]

Step 2: Calculate under 18s tickets: 5,062 − 3,484 = 1,578 tickets [M1]

Step 3: Calculate under 18s revenue: 1,578 × £12 = £18,936 [M1]

Step 4: Total revenue = £62,712 + £18,936 = £81,648 [A1]

Step 5: Charity donation (15%): £81,648 × 0.15 = £12,247.20 [A1]

£ 12 247.20

[5 marks]

(5 marks)

Q13 A gardener needs to buy compost to fill this tub.

The diagram shows the dimensions of the tub

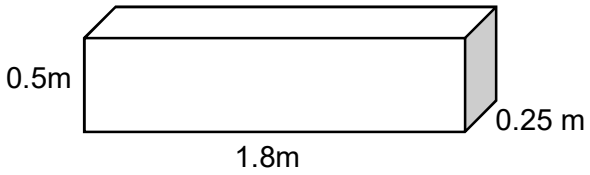


Diagram not to scale

They buy compost in 50 litre bags

1000 litres = 1m³

How many bags of compost must they buy?

Show all your working

Step 1: Calculate volume of tub in cubic metres (Volume = l × w × h):

1.8 m × 0.5 m × 0.25 m = 0.225 m³ [M1]

Step 2: Convert volume to litres (1 m³ = 1,000 litres):

0.225 × 1,000 = 225 litres needed [M1]

Step 3: Calculate number of 50-litre bags needed:

225 litres ÷ 50 litres = 4.5 bags [M1]

Must buy whole bags: round up to 5 bags [A2]

Number of bags 5

[5 marks]

(5 marks)

Q14 20 students take a maths test. Their tutor records the scores. The list below shows their scores:

15, 24, 6, 8, 29, 6, 12, 17, 24, 22, 25, 30, 18, 3, 11, 7, 18, 27, 28, 27

Present these scores in **three** suitable groups

Grouped Frequency Table (Three Equal Intervals):

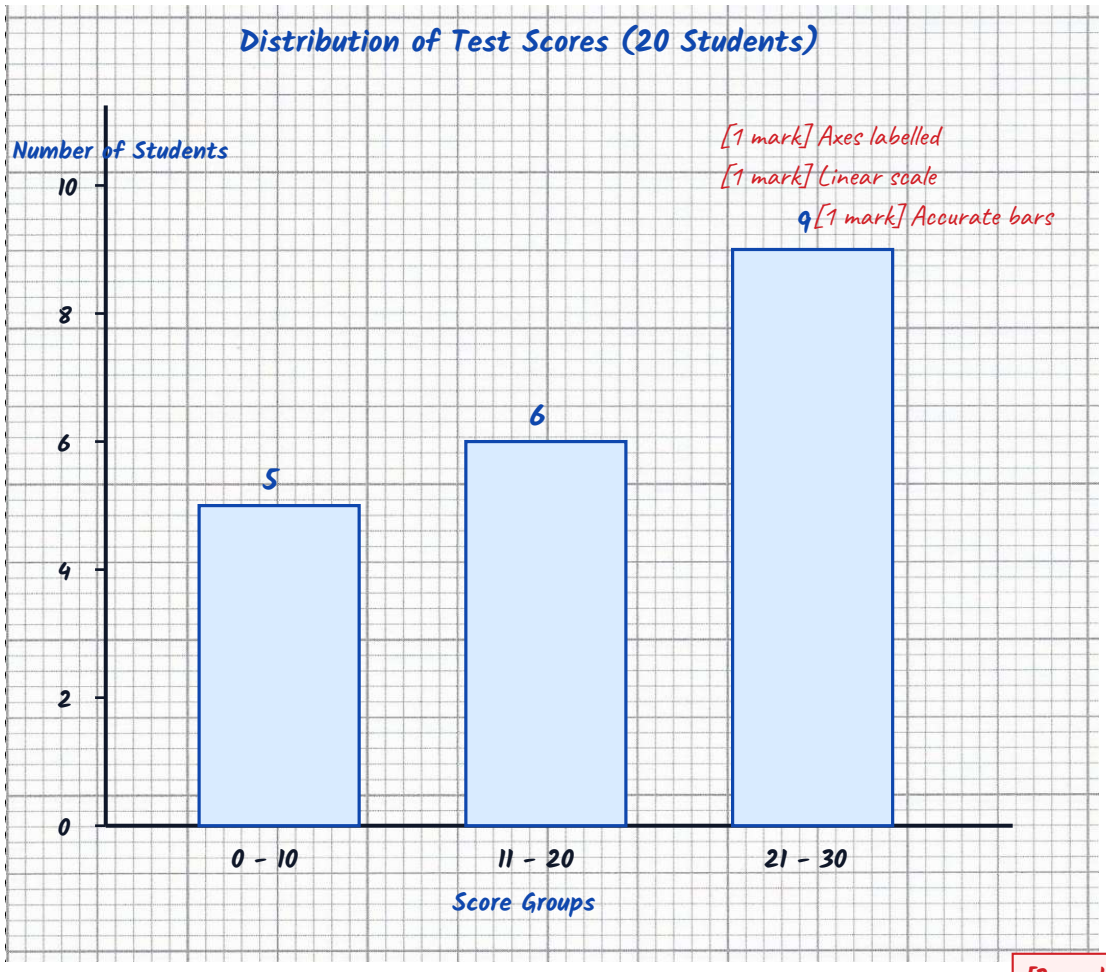
Score Group	Raw Scores	Frequency
0 – 10	3, 6, 6, 7, 8	5
11 – 20	11, 12, 15, 17, 18, 18	6
21 – 30	22, 24, 24, 25, 27, 27, 28, 29, 30	9

Total: 20 students

[2 marks]

(2 marks)

Present this information in a suitable chart or graph.



[3 marks]

(3 marks)

There is spare graph paper on pg 21

Q15 A school thinks their results are better than the national average.

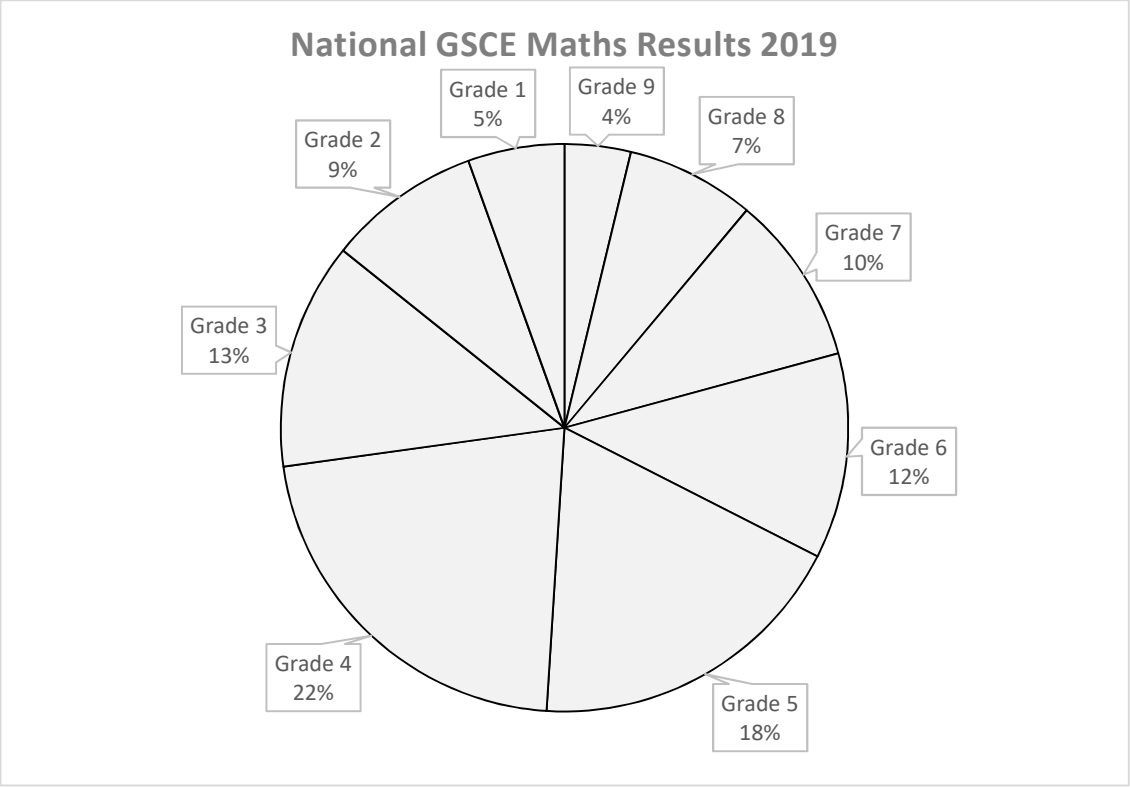
The school has a target for 65% of their students to achieve a Grade 4 or higher in their Maths GSCE.

A school governor uses this school table to check results.

School GSCE Maths Results 2019	
Grade	Number of Students
9 - Highest	7
8	11
7	18
6	22
5	27
4	55
3	27
2	22
1 - Lowest	11

She wants to compare her school's results with the national Maths GCSE results for Grade 4 and above.

She sees these national results.



She needs to write a report for the board of governors.

She says that the school has met its target **and** that its results are better than the national average.

Is the governor correct?
Explain your decisions. Show calculations to support your explanation.

Step 1: Calculate school students achieving Grade 4 or higher (from table):

Grade 4+ = 7 + 11 + 18 + 28 + 36 + 40 = 140 students out of 200 [M1]

Show all your working.

School percentage = (140 ÷ 200) × 100 = 70% [A1]

Step 2: Compare with school target (Target is 65%):

70% > 65% --> Target is met! [A1]

Step 3: Calculate national average for Grade 4 or higher (from pie chart):

National Grade 4+ = 4% + 7% + 10% + 12% + 18% + 22% = 73% [M1]

Target met? *(tick one box)* **Yes** ☑ **No** ☐

Better than national average? *(tick one box)* **Yes** ☐ **No** ☑

Explanation

Explanation & Conclusion:

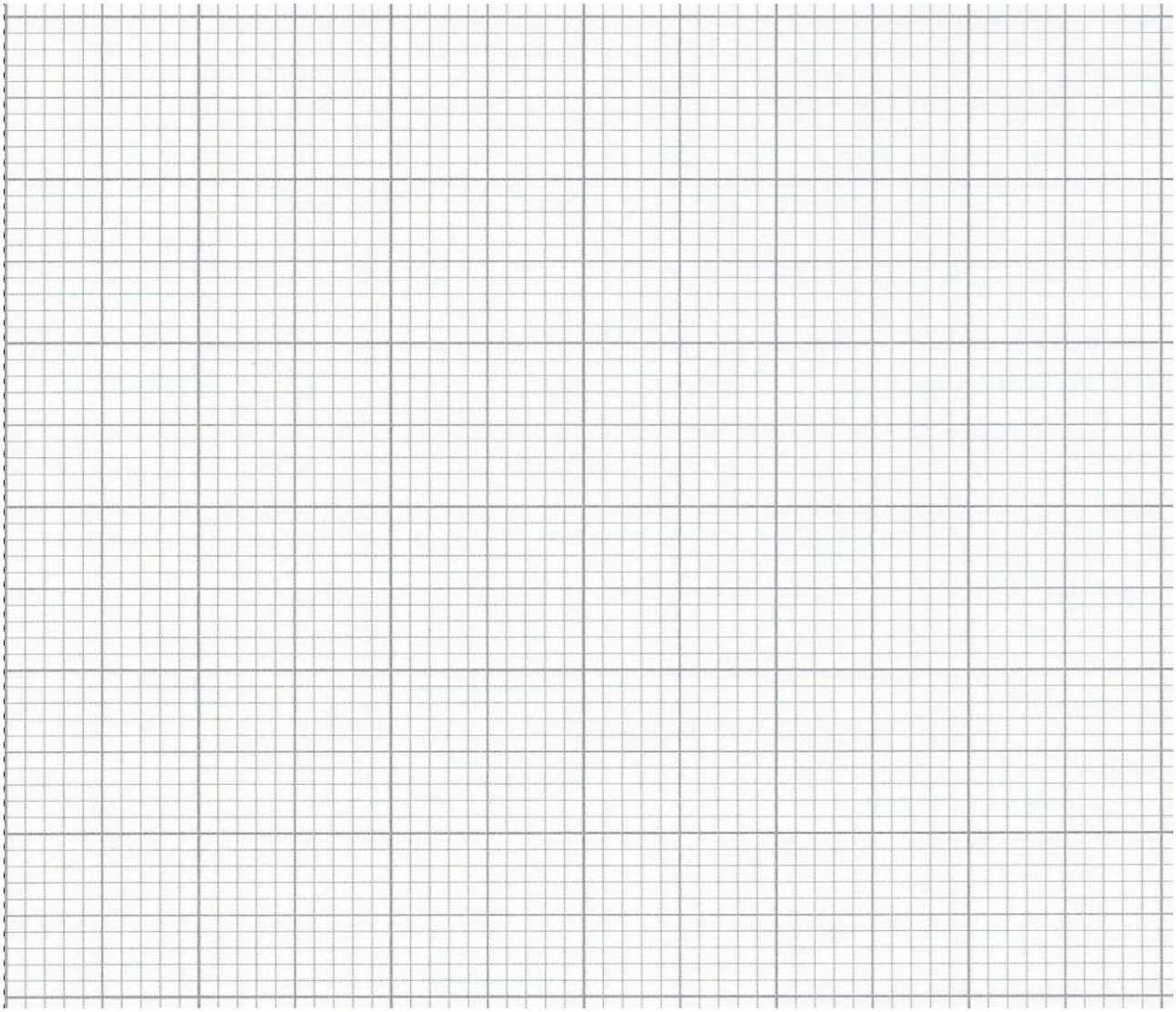
1. The school met its target because 70% achieved Grade 4+, which exceeds 65%.

2. The governor is INCORRECT because the school result (70%) is lower than the national average (73%). The school is 3% below national performance. [A2]

[6 marks]

(6 marks)

Spare graph paper for Question 14



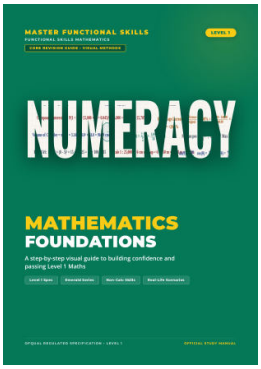
End of Section 2

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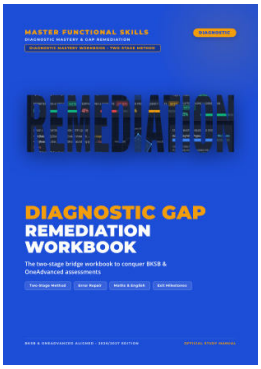


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